UnoLib documentation

stringutils.pas

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The stringutils.pas module contains routines for string handling. These routines can be used on all AVR microcontrollers supported by the Free Pascal Compiler. The module provides two sets of conversion routines: simplified functions that return a TIntString (11 characters), and more flexible functions that operate on PChar buffers.

Routines
`function` **`UInt8ToHexString`**`(Val: UInt8): TIntString;` Returns a hexadecimal representation of *Val* as TIntString.
`function` **`UInt16toHexString`**`(Val: UInt16): TIntString;` Returns a hexadecimal representation of *Val* as TIntString.
`function` **`UInt8ToString`**`(Val: UInt8): TIntString;` Converts *Val* to TIntString.
`function` **`Int8ToString`**`(Val: Int8): TIntString;` Converts *Val* to TIntString.
`function` **`UInt16ToString`**`(Val: UInt16): TIntString;` Converts *Val* to TIntString.
`function` **`Int16ToString`**`(Val: Int16): TIntString;` Converts *Val* to TIntString.
`function` **`UInt32ToString`**`(Val: UInt32): TIntString;` Converts *Val* to TIntString.
`function` **`Int32ToString`**`(Val: Int32): TIntString;` Converts *Val* to TIntString.
`function` **`UInt32Digits`**`(aVal: UInt32): UInt8;` Returns the number of decimal digits in a 32-bit unsigned integer (e.g. for 12345 returns 5).
`function` **`UInt16Digits`**`(aVal: UInt16): UInt8;` Returns the number of decimal digits in a 16-bit unsigned integer (e.g. for 1234 returns 4).
`function` **`UInt8Digits`**`(aVal: UInt8): UInt8;`

Returns the number of decimal digits in a 8-bit unsigned integer (e.g. for 123 returns 3).

```
function UInt32ToStr(const s: PChar; const maxlen, digits:
UInt8; const Val: UInt32): UInt8;
```

Converts an unsigned 32-bit integer number to a null-terminated string and stores it in a provided buffer. The function returns the number of characters in the resulting string.

Parameters
s: A pointer to a character buffer where the null-terminated string will be stored. The buffer should be declared as array[0..n] of char.
maxlen: The maximum length of the output string, including the null terminator. This value is typically the size of the *s* buffer.
digits: If this value is greater than number of decimal digits of *val* leading zeroes are added to the resulting string.
val: The 32-bit unsigned integer number to be converted.

Return value
The number of characters in the output string excluding null character. Returns 0 if the s buffer is too small to contain the converted string.

```
function Int32ToStr(const s: PChar; const maxlen, digits:
UInt8; const Val: Int32): UInt8;
```

Converts a signed 32-bit integer number to a null-terminated string and stores it in a provided buffer. The function returns the number of characters in the resulting string.

Parameters
s: A pointer to a character buffer where the null-terminated string will be stored. The buffer should be declared as array[0..n] of char.
maxlen: The maximum length of the output string, including the null terminator. This value is typically the size of the *s* buffer.
digits: If this value is greater than number of decimal digits of *val* leading zeroes are added to the resulting string.
val: The 32-bit signed integer number to be converted.

Return value
The number of characters in the output string excluding null character. Returns 0 if the s buffer is too small to contain the converted string.

```
function UInt16ToStr(const s: PChar; const maxlen, digits:
UInt8; const Val: UInt16): UInt8;
```

Converts an unsigned 16-bit integer number to a null-terminated string and stores it in a provided buffer. The function returns the number of characters in the resulting string.

Parameters
s: A pointer to a character buffer where the null-terminated string will be stored. The buffer should be declared as array[0..n] of char.
maxlen: The maximum length of the output string, including the null terminator. This value is typically the size of the *s* buffer.
digits: If this value is greater than number of decimal digits of *val* leading zeroes are added to

the resulting string.
val: The 16-bit unsigned integer number to be converted.

Return value
The number of characters in the output string excluding null character. Returns 0 if the s buffer is too small to contain the converted string.

```
function Int16ToStr(const s: PChar; const maxlen, digits: UInt8; const Val: Int16): UInt8;
```

Converts a signed 16-bit integer number to a null-terminated string and stores it in a provided buffer. The function returns the number of characters in the resulting string.

Parameters
s: A pointer to a character buffer where the null-terminated string will be stored. The buffer should be declared as array[0..n] of char.
maxlen: The maximum length of the output string, including the null terminator. This value is typically the size of the *s* buffer.
digits: If this value is greater than number of decimal digits of *val* leading zeroes are added to the resulting string.
val: The 16-bit signed integer number to be converted.

Return value
The number of characters in the output string excluding null character. Returns 0 if the s buffer is too small to contain the converted string.

```
function UInt8ToStr(const s: PChar; const maxlen, digits: UInt8; const Val: UInt16): UInt8;
```

Converts an unsigned 8-bit integer number to a null-terminated string and stores it in a provided buffer. The function returns the number of characters in the resulting string.

Parameters
s: A pointer to a character buffer where the null-terminated string will be stored. The buffer should be declared as array[0..n] of char.
maxlen: The maximum length of the output string, including the null terminator. This value is typically the size of the *s* buffer.
digits: If this value is greater than number of decimal digits of *val* leading zeroes are added to the resulting string.
val: The 8-bit unsigned integer number to be converted.

Return value
The number of characters in the output string excluding null character. Returns 0 if the s buffer is too small to contain the converted string.

```
function Int8ToStr(const s: PChar; const maxlen, digits: UInt8; const Val: Int16): UInt8;
```

Converts a signed 8-bit integer number to a null-terminated string and stores it in a provided buffer. The function returns the number of characters in the resulting string.

Parameters
s: A pointer to a character buffer where the null-terminated string will be stored. The buffer

should be declared as array[0..n] of char.
maxlen: The maximum length of the output string, including the null terminator. This value is typically the size of the *s* buffer.
digits: If this value is greater than number of decimal digits of *val* leading zeroes are added to the resulting string.
val: The 8-bit signed integer number to be converted.

Return value
The number of characters in the output string excluding null character. Returns 0 if the s buffer is too small to contain the converted string.

```
function UInt8ToHexStr(const s: PChar; const maxlen, digits:
UInt8; const Val: UInt16): UInt8;
```

Converts an unsigned 8-bit integer number to a null-terminated hexadecimal string and stores it in a provided buffer. The function returns the number of characters in the resulting string.

Parameters
s: A pointer to a character buffer where the null-terminated string will be stored. The buffer should be declared as array[0..n] of char.
maxlen: The maximum length of the output string, including the null terminator. This value is typically the size of the *s* buffer.
digits: If this value is greater than number of decimal digits of *val* leading zeroes are added to the resulting string.
val: The 8-bit unsigned integer number to be converted.

Return value
The number of characters in the output string excluding null character. Returns 0 if the s buffer is too small to contain the converted string.

```
function UInt16ToHexStr(const s: PChar; const maxlen, digits:
UInt8; const Val: UInt16): UInt8;
```

Converts an unsigned 16-bit integer number to a null-terminated hexadecimal string and stores it in a provided buffer. The function returns the number of characters in the resulting string.

Parameters
s: A pointer to a character buffer where the null-terminated string will be stored. The buffer should be declared as array[0..n] of char.
maxlen: The maximum length of the output string, including the null terminator. This value is typically the size of the *s* buffer.
digits: If this value is greater than number of decimal digits of *val* leading zeroes are added to the resulting string.
val: The 16-bit unsigned integer number to be converted.

Return value
The number of characters in the output string excluding null character. Returns 0 if the s buffer is too small to contain the converted string.

Example

```
var
  Buf: array[0..10] of Char;
begin
  UInt16ToStr(Buf, SizeOf(Buf), 5, 123);
  // Buf = '00123'
end;
```